KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3062 | 3 Credits: 3 | Program core | Source-grounded

# Textile Wet Processing

 Identify the chemicals used and pre-treatments given to cotton fibres.

OFFICIAL SOURCE DECLARATION

## How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3062 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 3062.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

## Official course information

**Course information**

| Item | Official information |
|---|---|
| Programme | Diploma in Textile Technology |
| Course code | 3062 |
| Course title | Textile Wet Processing |
| Semester | 3 Credits: 3 |
| Course category | Program core |
| Periods per week | 3 (L:2, T:1, P:0) Periods per semester: 45 |
| Periods per semester | 45 |

| Item | Official information |
| --- | --- |
| Credits | 3 |
| Prerequisite | See the official syllabus PDF |
| Assessment | Use the official assessment section reproduced below |

### Modules

4 official module sections detected.

### Topic entries

12 source-aligned topic entries retained.

### Study mode

Theory and application emphasis.

STUDY METHOD

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

# Objectives, outcomes and mapping

## Course objectives

-  Identify the chemicals used and pre-treatments given to cotton fibres.
-  To impart the knowledge of dyeing natural cellulosic & Viscose fibres using water-
- soluble& water-insoluble dyes.

**Course outcomes**

| CO | Official outcome | Study level |
|---|---|---|
| CO1 | 11 Understanding fabrics Discuss various methods of bleaching textile | See official syllabus |
| CO2 | materials & Cotton Mercerisation process. 10 Understanding and Mercerizing Discuss the process of dyeing Cotton/Viscose fibre | See official syllabus |
| CO3 | 10 Understanding using water-soluble dyes. Explain the process ofcotton/ viscose fibre dyeing | See official syllabus |
| CO4 | 12 Understanding with water insoluble dyes. | See official syllabus |

**CO–PO mapping evidence**

| Row | Extracted official mapping |
|---|---|
| CO1 | CO1 3 |
| CO2 | CO2 3 |
| CO3 | CO3 3 |
| CO4 | CO4 3 |

COVERAGE AUDIT

## Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

**Module coverage**

| Module | Official heading | Topic entries | Hours |
|---|---|---|---|
| Module 1 | Demonstrate the pre-treatments given to Cotton Grey fabrics | 3 | As specified |
| Module 2 | Use of various methods of bleaching textile materials & Cotton | 2 | As specified |
| Module 3 | dyes. | 4 | As specified |
| Module 4 | Execute cotton/ viscose fibre dyeing with water insoluble dyes. | 3 | As specified |

MODULE 1

# Demonstrate the pre-treatments given to Cotton Grey fabrics

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Demonstrate the pre-treatments given to Cotton Grey fabrics

- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for

know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for is an official topic in Module 1 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, know various chemicals used in textile wet 4 remembering processing discuss the preparation of cotton/ viscose for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

## 4 Understanding dyeing

4 Understanding dyeing is an official topic in Module 1 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding dyeing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding dyeing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Understanding dyeing means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-[illegible] 4 Understanding dyeing [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working**

Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working is an official topic in Module 1 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, differentiate singeing, desizing & scouring 3 understanding chemistry of wetting: definition of surface tension and contact angle - relationship of surface tension and contact angle with wetting of textile materials preparation of cotton/ viscose for dyeing: sequence of operations in wet processing of cotton/ viscose material. singeing: objectives - methods - hot plate, roller, gas singeing machines. de-sizing: objectives - methods - rot steeping - acid steeping, enzymatic de-sizing. scouring: objectives and importance - mechanism of scouring- types of kiers. features of a modern kiers and its working should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-□□ Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Demonstrate the pre-treatments given to Cotton Grey fabrics

            Memorize the chemistry of Wetting and
  M1.01     know various chemicals used in textile wet          4
Remembering
            processing
            Discuss the preparation of cotton/ Viscose for
  M1.02                                                         4
Understanding
            dyeing

  M1.03     Differentiate Singeing, Desizing & Scouring         3
Understanding

Contents:
Chemistry of Wetting: Definition of surface tension and contact angle -
Relationship of
surface tension and contact angle with wetting of textile materials
Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet
processing of
Cotton/ Viscose material.
Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines.
De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-
sizing.
Scouring: Objectives and importance - mechanism of scouring- Types of Kiers.
Features
of a modern Kiers and its working
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 2**

# Use of various methods of bleaching textile materials & Cotton

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

| **As specified** | **CO-linked** |
| --- | --- |
| official hours | see outcomes |

### Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: Use of various methods of bleaching textile materials & Cotton
- Official duration: As specified
- Official topic entries captured: 2
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and

2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and is an official topic in Module 2 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding materials use of continuous method of scouring and bleaching process & compare hypochlorite 4 understanding m2.02 and peroxide bleaching gain the knowledge on the process and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and means in the official course context |
| Study lens | Principle → components or stages → result → application |

| Aspect | What to prepare |
|---|---|
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble**

importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble is an official topic in Module 2 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, importance of mercerisation to implement the 4 understanding same in new situation. bleaching: objectives, types of bleaching agents-comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - method of bleaching with sodium hypochlorite. properties of hydrogen peroxide- strength of hydrogen peroxide in volume & percentage - bleaching with hydrogen peroxide (batch process).continuous scouring and bleaching by using 'j' box. compare hypochlorite and peroxide bleaching. mercerisation: objects of mercerizing. changes in the appearance of cotton by mercerizing - various factors affecting the process of mercerisation. working of a hank mercerizing machine. working of continuous cloth mercerizing machine define ban. implement the process of dyeing cotton/viscose fibre using water-soluble should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-□□ importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite’s in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using ‘J’ Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Use of various methods of bleaching textile materials & Cotton
            Mercerisation process.
            Classify the bleaching process given to cotton
  M2.01                                                        2
Understanding
            materials
            Use of continuous method of scouring and
            Bleaching process & compare hypochlorite            4
Understanding
  M2.02
            and peroxide bleaching
            Gain the knowledge on the process and
  M2.03     importance of Mercerisation to implement the        4
Understanding
            same in new situation.
            Series Test - I                                     1
Contents:
Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium
hypochlorite's in terms of their properties, chemical behaviour and application -
Method of
bleaching with Sodium hypochlorite.
Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume &
percentage -
Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and
bleaching by
using 'J' Box. Compare hypochlorite and peroxide bleaching.
Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

# dyes.

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

### Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

### Key facts

- Official module title: dyes.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## 2 Understanding application Execute the application of Direct dyes,

2 Understanding application Execute the application of Direct dyes, is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 2 Understanding application Execute the application of Direct dyes, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 2 understanding application execute the application of direct dyes, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 2 Understanding application Execute the application of Direct dyes, means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ 2 Understanding application Execute the application of Direct dyes, □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

## 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on

3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding reactive dyes on cotton/ viscose fibres execute the application of basic dyes on should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## 3 Understanding Cotton material

3 Understanding Cotton material is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 3 Understanding Cotton material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 3 understanding cotton material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 3 Understanding Cotton material means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-[illegible] 3 Understanding Cotton material [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

**Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material**

Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material is an official topic in Module 3 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.

- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, test the fastness properties of dyed material 2 understanding introduction to dyeing: define dyes - selection of dye - classification of dyes based on the method of application - terminologies related to dyes and dyeing - material liquor ratio, shade percentage, percentage of exhaustion, affinity, substantivity. general mechanism of dyeing. important natural dyes. direct dyes: method of application of direct dye on cotton and viscose. after - treatments given to direct dyed materials. reactive dyes: properties - types - method of application of reactive dyes on cotton & viscose method of application of basic dyes on cotton. fastness properties of dyed material should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-□□ Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material ▯▯▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯ definition/meaning ▯▯▯▯▯▯▯▯▯▯▯▯▯. ▯▯▯▯▯▯ ▯▯▯▯▯▯ ▯▯▯▯▯▯▯, steps, application ▯▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯ ▯▯▯▯▯▯. Technical terms English-▯ ▯▯▯▯▯ ▯▯▯▯▯▯▯▯▯▯▯.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
dyes.
             Classify dyes based on the method of
  M3.01                                                        2
Understanding
             application
             Execute the application of Direct dyes,
  M3.02                                                        3
Understanding
             Reactive dyes on Cotton/ Viscose fibres
             Execute the application of Basic dyes on
  M3.03                                                        3
Understanding
             Cotton material

  M3.04      Test the fastness properties of dyed material     2
Understanding

Contents:
Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes
based on
the method of application - Terminologies related to dyes and dyeing - Material
Liquor
ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity.
General
mechanism of dyeing.
Important Natural dyes.
Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After -
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

# Execute cotton/ viscose fibre dyeing with water insoluble dyes.

This module is presented from the official syllabus scope for Textile Wet Processing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**
official hours

**CO-linked**
see outcomes

## Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Execute cotton/ viscose fibre dyeing with water insoluble dyes.
- Official duration: As specified
- Official topic entries captured: 3
- The full source excerpt is preserved below for coverage checking.

## 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile

4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile is an official topic in Module 4 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 remembering viscose materials with water-insoluble dyes application of ingrain dyes on textile should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

## 4 Understanding materials

4 Understanding materials is an official topic in Module 4 of Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding materials using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding materials should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
| --- | --- |
| Definition/identity | What 4 Understanding materials means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 4 Understanding materials □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

**Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/**

Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ is an official topic in Module 4 of

Textile Wet Processing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, test the fastness properties of dyed material 4 understanding water-insoluble dyes: sulphur dyes - method of application on cotton - defects of sulphur dying and remedies. vat dyes - types - vatting - method of application on cotton/ viscose material ingrain dyes: azoic colours - dyeing – diazotisation - development of colour on cotton brief idea on application of oxidised dyes, mineral dyes and natural indigo on cotton. fastness properties of dyed material text /reference: t/r booktitle/author v.a. shenai -technology of textile processing (volume 1 to 5) mahajan t1 publishers ahmedabad. r2 s.p.mishra- fibre science and technology r s prayag – scouring, bleaching and mercerizing and dyeing of cellulose r3 fibres r4 e.r trotman - bleaching, dyeing and chemical technology of textile fibres r5 srivastava – recent process in textile bleaching and dyeing r6 r rchakravarthi – a glimpse of the chemical technology of textile fibres r7 nalankilli.g., chemical preparatory process, ncute publication, new delhi online resources: sl.no website link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 october 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

**Topic study guide**

| Aspect | What to prepare |
|---|---|
| Definition/identity | What Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ means in the official course context |
| Study lens | Principle → components or stages → result → application |
| Exam evidence | Definition, labelled explanation, comparison, procedure or example as appropriate |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-[illegible] Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/ □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

### Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Execute cotton/ viscose fibre dyeing with water insoluble dyes.

             Execute the process of dyeing cotton and
  M4.01                                                          4
Remembering
             viscose materials with Water-insoluble dyes
             Application of Ingrain dyes on textile
  M4.02                                                          4
Understanding
             materials
  M4.03      Test the fastness properties of dyed material       4
Understanding

             Series Test - II                                    1
 Contents:
 Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects
of
 sulphur dying and remedies.
 Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material
 Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on
Cotton
 Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on
Cotton.
 Fastness properties of dyed material

Text /Reference:
```

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

## Concept and formula bank

### Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

### Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

## Diagram and source-transcript library

### Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

### Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

## Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

## Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

> **Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

**Define or explain know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for. (3 marks)**

**Answer:** Begin with the standard definition or identity of *know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding dyeing. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding dyeing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Differentiate Singeing, Desizing & Scouring 3 Understanding Chemistry of Wetting: Definition of surface tension and contact angle - Relationship of surface tension and contact angle with wetting of textile materials Preparation of Cotton/ Viscose for Dyeing: Sequence of operations in wet processing of Cotton/ Viscose material. Singeing: Objectives - Methods - Hot plate, Roller, Gas singeing machines. De-sizing: Objectives - Methods - Rot steeping - Acid steeping, Enzymatic de-sizing. Scouring: Objectives and importance - mechanism of scouring- Types of Kiers. Features of a modern Kiers and its working*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 2

**Define or explain 2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□□ definition, □□□□□□□ principle/steps, □□□□□□□ application □□□□□ □□□□□□□□□□□ □□□□□□□.

**Define or explain importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN. Implement the process of dyeing Cotton/Viscose fibre using water-soluble. (3 marks)**

**Answer:** Begin with the standard definition or identity of *importance of Mercerisation to implement the 4 Understanding same in new situation. Bleaching: Objectives, Types of bleaching agents-Comparison of sodium and calcium hypochlorite's in terms of their properties, chemical behaviour and application - Method of bleaching with Sodium hypochlorite. Properties of hydrogen peroxide- Strength of hydrogen peroxide in volume & percentage - Bleaching with Hydrogen peroxide (Batch process).Continuous scouring and bleaching by using 'J' Box. Compare hypochlorite and peroxide bleaching. Mercerisation: Objects of mercerizing. Changes in the appearance of cotton by mercerizing - Various factors affecting the process of Mercerisation. Working of a Hank Mercerizing machine. Working of continuous Cloth Mercerizing machine Define BAN.*

*Implement the process of dyeing Cotton/Viscose fibre using water-soluble*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 3

**Define or explain 2 Understanding application Execute the application of Direct dyes,. (3 marks)**

**Answer:** Begin with the standard definition or identity of *2 Understanding application Execute the application of Direct dyes,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding Reactive dyes on Cotton/ Viscose fibres Execute the application of Basic dyes on*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 3 Understanding Cotton material. (3 marks)**

**Answer:** Begin with the standard definition or identity of *3 Understanding Cotton material*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Test the fastness properties of dyed material 2 Understanding Introduction to Dyeing: Define dyes - selection of dye - Classification of dyes based on the method of application - Terminologies related to dyes and dyeing - Material Liquor ratio, Shade Percentage, Percentage of Exhaustion, affinity, substantivity. General mechanism of dyeing. Important Natural dyes. Direct Dyes: Method of application of Direct Dye on Cotton and Viscose. After - treatments given to Direct dyed materials. Reactive Dyes: Properties - Types - method of application of Reactive Dyes on Cotton & Viscose Method of application of Basic dyes on cotton. Fastness properties of dyed material*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

## Module 4

**Define or explain 4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain 4 Understanding materials. (3 marks)**

**Answer:** Begin with the standard definition or identity of *4 Understanding materials*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

**Define or explain Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/. (3 marks)**

**Answer:** Begin with the standard definition or identity of *Test the fastness properties of dyed material 4 Understanding Water-Insoluble dyes: Sulphur Dyes - Method of application on cotton - Defects of sulphur dying and remedies. Vat Dyes - Types - Vatting - Method of application on Cotton/ Viscose material Ingrain Dyes: Azoic Colours - Dyeing – Diazotisation - Development of Colour on Cotton Brief idea on Application of Oxidised dyes, Mineral dyes and Natural Indigo on Cotton. Fastness properties of dyed material Text /Reference: T/R BookTitle/Author V.A. Shenai -Technology of Textile Processing (Volume 1 to 5) Mahajan T1 publishers Ahmedabad. R2 S.P.Mishra- Fibre science and technology R S Prayag – Scouring, Bleaching and Mercerizing and Dyeing of cellulose R3 fibres R4 E.R Trotman - Bleaching, Dyeing and Chemical Technology of Textile fibres R5 Srivastava – Recent process in Textile Bleaching and Dyeing R6 R RChakravarthi – A Glimpse of the chemical technology of Textile fibres R7 Nalankilli.G., Chemical Preparatory Process, NCUTE Publication, New Delhi Online resources: Sl.No Website Link http://textlnfo.wordpress.com/2011/10/24/classification-of-textile-fibers- 1 according/ (accessed on 20 October 2013) 2 https://nptel.ac.in/courses/116/102/116102026/ 3 https://nptel.ac.in/courses/116/102/116102016/ 4 https://swayam.gov.in/ 5 http://eacharya.inflibnet.ac.in/vidya-mitra/*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **know various chemicals used in textile wet 4 Remembering processing Discuss the preparation of cotton/ Viscose for**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **2 Understanding materials Use of continuous method of scouring and Bleaching process & compare hypochlorite 4 Understanding M2.02 and peroxide bleaching Gain the knowledge on the process and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **2 Understanding application Execute the application of Direct dyes,**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 4

1-mark definition: State the meaning of **4 Remembering viscose materials with Water-insoluble dyes Application of Ingrain dyes on textile**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

## Exam checklist

### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

## Glossary

**Study glossary**

| Term | Meaning in this lesson |
|---|---|
| Course Outcome | A capability the student should demonstrate after completing the course. |
| Module | An official syllabus unit with defined topics and hours. |
| CIE | Continuous Internal Evaluation. |
| ESE | End Semester Examination. |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

SOURCE-SUPPORTED REFERENCES

## References and web resources

### References listed in the official PDF

References are included in the official source PDF.

### Web resources listed in the official PDF

- Sl.No Website Link
- http://textInfo.wordpress.com/2011/10/24/classification-of-textile-fibers- according/ (accessed on 20 October 2013)
- https://nptel.ac.in/courses/116/102/116102026/
- https://nptel.ac.in/courses/116/102/116102016/ https://swayam.gov.in/ http://eacharya.inflibnet.ac.in/vidya-mitra/

IN-PAGE SEARCH

## Search this lesson